Alejandro B. Kolton

Physicist — non-equilibrium disordered systems and computational statistical physics

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Positions

2022– **Professor (Profesor Adjunto), Instituto Balseiro**
Universidad Nacional de Cuyo (UNCUYO), Bariloche, Argentina.

2007– **Principal Researcher, CONICET**
Condensed Matter Theory group, Centro Atómico Bariloche, Argentina. Permanent researcher since 10/2007.

2006–2007 **Postdoctoral researcher, Universidad Complutense de Madrid**
J. M. R. Parrondo group, Dep. de Física Atómica, Molecular y Nuclear, Spain.

2003–2006 **Postdoctoral researcher and Maître assistant, Université de Genève**
T. Giamarchi group, DPMC, Switzerland.

Education

2003 **PhD in Physics, Instituto Balseiro**
Thesis: *Nonequilibrium dynamics in vortex systems in superconductors*.
2nd J. J. Giambiagi Prize for the PhD thesis, Asociación Física Argentina.

1998 **Master in Physics (Licenciatura), Instituto Balseiro**

Visiting positions

2018 **LPTMS, Université Paris-Saclay, France**

2016 **LiPhy, Université Grenoble Alpes, France**

2013 **École Normale Supérieure, Paris, France**

2011 **Dept. de Física Fonamental, Universitat de Barcelona, Spain**

Research interests

Driven extended systems in disordered landscapes: domain walls in ferromagnets and ferroelectrics, vortices in superconductors, magnetic skyrmions, amorphous solids, earthquakes, electronic glasses, cold atoms, propagation of epidemics and forest fires, active matter. Disordered elastic systems, out-of-equilibrium glassy dynamics, stochastic thermodynamics, computational statistical physics and GPU computing.

Publications

Over 90 papers in international journals, including Phys. Rev. Lett., Phys. Rev. B/E, J. Stat. Mech., Communications Physics and Communications Materials. Google Scholar: 3331 citations, h-index 34.
Full, up-to-date list: alejandro-kolton.github.io/publications.html

Selected publications

- Depinning free of the elastic approximation
  A. B. Kolton, E. E. Ferrero, A. Rosso — *Phys. Rev. B 108, 174201 (2023)*
- Universal pinning energy barrier for driven domain walls in thin ferromagnetic films
  V. Jeudy, A. Mougin, S. Bustingorry, W. Savero Torres, J. Gorchon, A. B. Kolton, A. Lemaître, J.-P. Jamet — *Phys. Rev. Lett. 117, 057201 (2016)*
- Creep dynamics of elastic manifolds via exact transition pathways
  A. B. Kolton, A. Rosso, T. Giamarchi, W. Krauth — *Phys. Rev. B 79, 184207 (2009)*

- Mode locking in ac-driven vortex lattices with random pinning
  A. B. Kolton, D. Domínguez, N. Grønbech-Jensen — *Phys. Rev. Lett. 86, 4112 (2001)*
- Hall noise and transverse freezing in driven vortex lattices
  A. B. Kolton, D. Domínguez, N. Grønbech-Jensen — *Phys. Rev. Lett. 83, 3061 (1999)*

Patent. *Soporte magnético para la grabación y lectura de información y su uso*, A. Pérez-Junquera, V. I. Marconi, A. B. Kolton *et al.*, ES2406182 A2 (2013).

Recent invited talks

2026 **Avalanche 2026 workshop, Anacapri, Italy**
Universal collective creep and avalanches of magnetic domain walls: bridging theory, numerics, and experiment.

2026 **ICTP-SAIFR Workshop on Non-equilibrium Statistical Physics, São Paulo, Brazil**
When driven interfaces go wild: chaos and roughening without disorder.

2026 **LAFMAT, Latin American Network of Condensed Matter Physics and Materials Science**
Spatio-temporal chaos in field-driven magnetic domain walls.

More talks and slides: alejandro-kolton.github.io/talks.html

Teaching

- Instituto Balseiro: Statistical Mechanics; Introduction to Numerical Computing on Graphics Processors (GPUs).
- Invited teacher, Erasmus Mundus Master in Complex Systems Science, Université Paris Diderot (2015).
- Maître assistant, Université de Genève (2003–2006).

Supervision

Current: E. F. Pusiol (PhD, 2025–); A. M. Cámara (PhD, co-advisor, 2025–); M. E. Romano (PhD, co-advisor with J. Curiale, 2026–).

Past: P. Domenichini (postdoc, 2022–2025); A. Gonzalez (PhD, 2023–2024); F. Elías (PhD, 2017–2023); V. H. Purrello (PhD, 2015–2020); E. E. Ferrero (postdoc, 2011–2013). All CONICET fellowships.

Service and outreach

- Chair, *Yielding Phenomena in Disordered Systems* conference, Instituto Balseiro (2019).
- Chair, *Third Argentinean School of GPGPU Computing for Scientific Applications*, Instituto Balseiro (2014).
- Low-cost CO_2 meters to reduce indoor COVID-19 contagion risk: open hardware design with pedagogical instructions for technical high schools, Instituto Balseiro extension project (2021). github.com/droyktton/medidorCO2

Languages

Spanish (native), English, French.